Procedural Adaptive Footstep System
Detailed Jira-Style Task Plan

EPIC 1 — Core Planning & Architecture

AUD-001 — Define Procedural Footstep Design

- List all footstep layers (impact, body, clothing).
- Define signal flow from trigger to output.
- Decide which parameters affect each layer.
- Document expected behavior per parameter.

AUD-002 — System Architecture & Data Flow

- Define core classes and responsibilities.
- Decide ownership of audio resources.
- Design data flow between parameter state and audio engine.
- Ensure UI is isolated from audio logic.

AUD-003 — JUCE Project Setup

- Create new JUCE application project.
- Configure audio device settings.
- Verify audio callback execution.
- Confirm silence output without errors.

EPIC 2 — Parameter System

AUD-004 — Gameplay Parameter State

- Create a shared parameter struct or class.
- Implement atomic or lock-free access.
- Add smoothing for continuous parameters.
- Expose setters for UI control.

AUD-005 — Footstep Timing Logic

- Implement timer or sample-based clock.
- Map speed parameter to step interval.
- Trigger footstep events deterministically.
- Test stability under rapid changes.

EPIC 3 — Procedural DSP Building Blocks

AUD-006 — Noise Generator

- Implement white noise generation.
- Add gain control.
- Ensure no allocations in callback.
- Validate signal stability.

AUD-007 — Envelope Generator

- Design envelope shape (attack/decay).

- Trigger envelope per footstep.
- Scale envelope time with speed.
- Prevent clicks and pops.

AUD-008 — Parameter-to-DSP Mapping

- Map speed to envelope and pitch.
- Map weight to gain and filtering.
- Clamp values to safe ranges.
- Test transitions for smoothness.

EPIC 4 — Impact Layer

AUD-009 — Impact Sample Loader

- Organize samples per surface.
- Load samples at startup.
- Handle missing or invalid files.
- Store samples in memory safely.

AUD-010 — Impact Playback Voice

- Implement one-shot playback.
- Apply pitch modulation from speed.
- Apply gain scaling from weight.
- Ensure voice stops cleanly.

EPIC 5 — Body Layer

AUD-011 — Procedural Body Layer

- Generate noise-based texture.
- Apply envelope shaping.
- Filter based on surface type.
- Emphasize low frequencies with weight.

EPIC 6 — Clothing Foley Layers

AUD-012 — Clothing Layer Architecture

- Create modular clothing layer class.
- Allow enabling/disabling layers.
- Assign unique modulation per layer.
- Integrate with main trigger.

AUD-013 — Procedural Cloth Sound

- Design noise-based cloth sound.
- Modulate amplitude with speed.
- Add random variation per step.
- Balance against other layers.

EPIC 7 — Layer Mixing & Voice Control

AUD-014 — Footstep Layer Mixer

- Sum all layers into final output.
- Apply gain staging.
- Prevent clipping.
- Test mix balance.

AUD-015 — Voice Lifecycle Management

- Trigger all layers simultaneously.
- Stop layers when envelopes finish.
- Prevent lingering audio.
- Handle rapid retriggers.

EPIC 8 — UI & Debug Controls

AUD-016 — Debug Control UI

- Create sliders for speed and weight.
- Add surface selector.
- Add clothing layer toggles.
- Verify real-time response.

EPIC 9 — Stability & Optimization

AUD-017 — Real-Time Safety Pass

- Audit audio callback for allocations.
- Remove locks or blocking calls.
- Measure CPU usage.
- Fix performance issues.

AUD-018 — Edge Case Handling

- Test extreme parameter values.
- Handle missing data safely.
- Ensure silence when disabled.
- Prevent crashes.

EPIC 10 — Documentation & Portfolio

AUD-019 — Technical Documentation

- Document architecture.
- Explain parameter mappings.
- Describe DSP choices.
- Add build instructions.

AUD-020 — Portfolio Write-Up

- Write project overview.
- Explain game audio relevance.
- Include diagrams or screenshots.
- Prepare for public presentation.